

MYMENSINGH GIRLS' CADET COLLEGE
SECOND TERM-END EXAMINATION - 2026
CLASS: XII
MULTIPLE CHOICE QUESTIONS
STATISTICS
SECOND PAPER
[According to the Syllabus of 2027]
TIME – 25 minutes
FULL MARKS – 25

Subject Code:

1	3	0
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Set:

Ka

[N.B. – Answer all the questions. Each question carries ONE mark. Block fully, with a black ball- point pen, the circle of the letter that stands for the correct/best answer in the “Answer sheet” for the Multiple Choice Questions Examination.]

Candidates are asked not to leave any mark or spot on the question paper.

1. What is the probability of a certain event (an event guaranteed to occur)?

(a) 0 (b) 0.5
2. If two events A and B are mutually exclusive (disjoint), which of the following is true?

(a) $P(A \cap B) = P(A)P(B)$
(c) $P(A \cup B) = 1$
3. Which of the following statements regarding probability are correct?

i. $0 \leq P(A) \leq 1$ for any event A
ii. $P(S) = 1$ where S is the sample space
iii. For mutually exclusive events A and B , $P(A \cup B) = P(A) + P(B)$

Which one is correct?

(a) i and ii (b) i and iii
4. Which of the following are necessary conditions for a discrete probability function $P(x)$?

(c) 1 (d) Undefined

i. $P(x_i) \geq 0$ for all x_i
ii. $\sum P(x_i) = 1$
iii. $P(x_i) \leq 1$ for all x_i

Which one is correct?

(a) i and ii (b) i and iii (c) ii and iii
5. Answer the next TWO questions based on the following information

A discrete random variable X has the following probability distribution:

X	0	1	2
$P(x)$	0.50	0.25	0.25
6. What is the value of the cumulative distribution function $F(1)$?

(a) 0.25 (b) 0.50 (c) 0.75 (d) 1.00
7. What is $P(X \geq 1)$?

(a) 0.25 (b) 0.50 (c) 0.75 (d) 1.00
8. If X is a random variable and c is a constant, what is the variance $V(c)$?

(a) c (b) c^2 (c) 0 (d) 1
9. If $E(X) = 4$ and $V(X) = 5$, what is the value of $E(X^2)$?

(a) 9 (b) 16 (c) 29 (d) 39
10. If $Y = 3X + 5$ and $V(X) = 4$, what is the variance $V(Y)$?

(a) 17 (b) 36 (c) 49 (d) 61
11. For two independent random variables X and Y , which of the following properties are true?

(b) Height of students in a class
(d) Outcome of rolling a die

i. $E(X + Y) = E(X) + E(Y)$
ii. $E(XY) = E(X)E(Y)$
iii. $V(X - Y) = V(X) + V(Y)$

Which one is correct?

(c) 0 (a) i and ii (d) 1 (b) i and iii (c) ii and iii

Answer the next TWO questions based on the following information

In a group of 100 students, 60 passed in Mathematics, 50 passed in Statistics, and 30 passed in both subjects. A student is selected at random.

4. What is the probability that the student passed in Mathematics or Statistics?
- (a) 0.80 (b) 0.50
5. What is the conditional probability that the student passed in Statistics given that s/he passed in Mathematics?
- (a) 0.30 (b) 0.50
6. Which of the following is an example of a continuous random variable?
- (a) Number of defective items in a batch
(c) Number of phone calls received in an hour
7. What is $F(\infty)$ for any valid cumulative distribution function (CDF) $F(x)$?
- (a) $-\infty$ (b) -1

15. If $E(X) = 5$, what is the value of $E(2X - 3)$?

(a) 7

(b) 10

16. The first central moment (μ_1) of any frequency distribution is always equal to –

(a) 1

(b) 0

17. Which moment is used to measure the variance (dispersion) of a distribution?

(a) First raw moment (μ'_1)

(c) Second central moment (μ_2)

18. For a mesokurtic (normal-shaped) distribution, what is the value of kurtosis β_2 ?

(a) 0

(b) 1

19. Which of the following statements about moments are correct?

i. The first raw moment around zero (μ'_1) is the arithmetic mean.

ii. The third central moment (μ_3) can be negative, zero, or positive.

iii. The second central moment (μ_2) can never be negative.

Which one is correct?

(a) i and ii

(b) i and iii

20. If the first raw moment about 4 is -2 , what is the value of the arithmetic mean?

(a) 2

(b) -2

21. The correlation coefficient r between two variables lies within which range?

(a) $0 \leq r \leq 1$

(b) $-1 \leq r \leq 1$

(c) $-1 \leq r \leq 0$

(d) $0 \leq r \leq 1$

22. If the correlation coefficient between X and Y is $r = 0.6$, what is the correlation coefficient between $2X + 3$ and $3Y - 5$?

(a) 0.6

(b) 1.2

(c) 0

(d) -1

23. The correlation coefficient r is the geometric mean of –

(a) Two means

(b) Two standard deviations

(c) Two regression coefficients (b_{yx} and b_{xy})

(d) -3

24. Which of the following statements are true about the correlation coefficient r ?

i. It is independent of change of origin and scale.

ii. It is a unitless measure.

iii. Correlation implies a causal relationship between variables.

Which one is correct?

(a) i and ii

(b) i and iii

(c) ii and iii

(d) i, ii and iii

25. If the two regression coefficients are $b_{yx} = 0.8$ and $b_{xy} = 0.45$, what is the value of the correlation coefficient r ?

(a) 0.32

(b) 0.60

(c) 0

(d) 4

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[N.B. – Answer all the questions. Each question carries ONE mark. Block fully, with a black ball- point pen, the circle of the letter that stands for the correct/best answer in the “Answer sheet” for the Multiple Choice Questions Examination.]

Candidates are asked not to leave any mark or spot on the question paper.

1. If $Y = 3X + 5$ and $V(X) = 4$, what is the variance $V(Y)$?

(a) 17

(b) 36

(c) 41

(d) 12
2. For two independent random variables X and Y , which of the following properties are true?

i. $E(X + Y) = E(X) + E(Y)$

ii. $E(XY) = E(X)E(Y)$

iii. $V(X - Y) = V(X) + V(Y)$

Which one is correct?

(a) i and ii

(b) i and iii

(c) ii and iii

(d) i, ii and iii

3. If $E(X) = 5$, what is the value of $E(2X - 3)$?

(a) 7

(b) 10

(c) 13

(d) 5

4. The first central moment (μ_1) of any frequency distribution is always equal to –

(a) 1

(b) 0

(c) -1

(d) Mean

5. Which moment is used to measure the variance (dispersion) of a distribution?

(a) First raw moment (μ'_1)

(b) First central moment (μ_1)

(c) Second central moment (μ_2)

(d) Third central moment (μ_3)

6. For a mesokurtic (normal-shaped) distribution, what is the value of kurtosis β_2 ?

(a) 0

(b) 1

(c) 3

(d) -3

7. Which of the following statements about moments are correct?

i. The first raw moment around zero (μ'_1) is the arithmetic mean.

ii. The third central moment (μ_3) can be negative, zero, or positive.

iii. The second central moment (μ_2) can never be negative.

Which one is correct?

(a) i and ii

(b) i and iii

(c) ii and iii

(d) i, ii and iii

8. If the first raw moment about 4 is -2 , what is the value of the arithmetic mean?

(a) 0.6

(b) 1.2

(c) 0

(d) 1.8

9. What is the probability of a certain event (an event guaranteed to occur)?

(a) 0

(b) 0.5

(c) 1

(d) 0.1

10. If two events A and B are mutually exclusive (disjoint), which of the following is true?

(a) $P(A \cap B) = P(A)P(B)$

(b) $P(A \cup B) = P(A) + P(B)$

(c) $P(A \cup B) = 1$

(d) $P(A \cap B) = 1$

11. Which of the following statements regarding probability are correct?

i. $0 \leq P(A) \leq 1$ for any event A

ii. $P(S) = 1$ where S is the sample space

iii. For mutually exclusive events A and B , $P(A \cup B) = P(A) + P(B)$

Which one is correct?

(a) i and ii

(b) i and iii

(c) ii and iii

(d) i, ii and iii

12. What is the probability that the student passed in Mathematics or Statistics?

(a) 0.80

(b) 0.50

(c) 0

(d) -3

13. What is the conditional probability that the student passed in Statistics given that s/he passed in Mathematics?

(a) 0.30

(b) 0.50

(c) 0

(d) 1

14. The correlation coefficient r between two variables lies within which range?

(a) $0 \leq r \leq 1$

(b) $-1 \leq r \leq 1$

(c) $-1 < r < 1$

(d) $-1 \leq r < 1$

15. If the correlation coefficient between X and Y is 0.6 , what is the correlation coefficient between $2X + 3$ and $3Y - 5$?

(a) 0.6

(b) 1.2

(c) 0

(d) 1.8

Answer the next TWO questions based on the following information

In a group of 100 students, 60 passed in Mathematics, 50 passed in Statistics, and 30 passed in both subjects. A student (s/he) is selected at random.

16. The correlation coefficient r is the geometric mean of –

(a) Two means

(c) Two regression coefficients (b_{yx} and b_{xy})

i. $P(x_i) \geq 0$ for all x_i

ii. $\sum P(x_i) = 1$

iii. $P(x_i) \leq 1$ for all x_i

(b) Two variances

(d) Two standard deviations
- Which one is correct?
- (a) i and ii

(b) i and iii

(c) ii and iii
17. Which of the following statements are true about the correlation coefficient r ?

i. It is independent of change of origin and scale.

ii. It is a unitless measure.

iii. Correlation implies a causal relationship between variables.

Which one is correct?

(a) i and ii

(b) i and iii

(c) ii and iii
- Answer the next TWO questions based on the following information

A discrete random variable X has the following probability distribution:

X	0	1	2
$P(x)$	0.50	0.25	0.25
18. If the two regression coefficients are $b_{yx} = 0.8$ and $b_{xy} = 0.45$, what is the value of the correlation coefficient r ?

(a) 0.32

(b) 0.60

(c) 0.70
22. What is the value of the cumulative distribution function $F(1)$?

(a) 0.25

(b) 0.50

(c) 0.75

(d) 0.80
19. Which of the following is an example of a continuous random variable?

(a) Number of defective items in a batch

(c) Number of phone calls received in an hour
23. What is $P(X \geq 1)$?

(a) 0.25

(b) 0.50

(c) 0.75

(d) 0.80
20. What is $F(\infty)$ for any valid cumulative distribution function (CDF) $F(x)$?

(a) $-\infty$

(b) -1

(c) 0

(d) 1
24. If X is a random variable and c is a constant, what is the variance $V(c)$?

(a) c

(b) c^2

(c) 0

(d) 1
21. Which of the following are necessary conditions for a discrete probability function $P(x)$?
25. If $E(X) = 4$ and $V(X) = 5$, what is the value of $E(X^2)$?

(a) 9

(b) 16

(c) 20

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Candidates are asked not to leave any mark or spot on the question paper.

1. Which of the following is an example of a continuous random variable?
(a) Number of defective items in a batch
(c) Number of phone calls received in an hour

2. What is $F(\infty)$ for any valid cumulative distribution function (CDF) $F(x)$?
(a) $-\infty$
(b) -1

3. Which of the following are necessary conditions for a discrete probability function $P(x)$?
i. $P(x_i) \geq 0$ for all x_i
ii. $\sum P(x_i) = 1$
iii. $P(x_i) \leq 1$ for all x_i
Which one is correct?
(a) i and ii
(b) i and iii

Answer the next TWO questions based on the following information
A discrete random variable X has the following probability distribution:

X	0	1	2
$P(x)$	0.50	0.25	0.25

4. What is the value of the cumulative distribution function $F(1)$?
(a) 0.25
(b) 0.50

5. What is $P(X \geq 1)$?
(a) 0.25
(b) 0.50

6. The correlation coefficient r between two variables lies within which range?
(a) $0 \leq r \leq 1$
(b) $-1 \leq r \leq 1$

7. If the correlation coefficient between X and Y is $r = 0.6$, what is the correlation coefficient between $2X + 3$ and $3Y - 5$?
(a) 0.6
(b) 1.2

8. The correlation coefficient r is the geometric mean of –
(b) Height of two students in a class
(c) Two regression coefficients (b_{yx} and b_{xy})
(d) Outcome of tossing a coin

9. Which of the following statements are true about the correlation coefficient r ?
(c) 0
i. It is independent of change of origin and scale.
ii. It is a unitless measure.
iii. Correlation implies a causal relationship between variables.
Which one is correct?
(a) i and ii
(b) i and iii
(c) ii and iii

10. If the two regression coefficients are $b_{yx} = 0.8$ and $b_{xy} = 0.45$, what is the value of the correlation coefficient r ?
(c) ii and iii
(a) 0.32
(d) i, ii and iii
(b) 0.60
(c) 0

11. The first central moment (μ_1) of any frequency distribution is always equal to –
(a) 1
(b) 0
(c) -1

12. Which moment is used to measure the variance (dispersion) of a distribution?
(a) First raw moment (μ'_1)
(c) Second central moment (μ_2)
(b) First central moment (μ_1)
(d) Third central moment (μ_3)

13. For a mesokurtic (normal-shaped) distribution, what is the value of kurtosis β_2 ?
(a) 0
(c) 0.75
(b) 1
(d) 1.00

14. Which of the following statements about moments are correct?
i. The first raw moment around zero (μ'_1) is the arithmetic mean.
ii. The third central moment (μ_3) can be negative, zero, or positive.
iii. The second central moment (μ_2) can never be negative.
Which one is correct?
(c) 0.2
(a) i and ii
(d) Cannot be determined
(b) i and iii
(c) ii and iii

15. If the first raw moment about 4 is -2 , what is the value of the arithmetic mean?

(a) 2 (b) -2

20. What is the conditional probability that the student passed in Statistics given that s/he passed in Mathematics?

(a) 0.30 (b) 0.50 (c) 0 (d) 4
16. What is the probability of a certain event (an event guaranteed to occur)?

(a) 0 (b) 0.5
21. If X is a random variable and c is a constant, what is the variance $V(c)$?

(a) c (b) c^2 (c) 0 (d) Undefined
17. If two events A and B are mutually exclusive (disjoint), which of the following is true?

(a) $P(A \cap B) = P(A)P(B)$

(c) $P(A \cup B) = 1$
22. If $E(X) = 4$ and $V(X) = 5$, what is the value of $E(X^2)$?

(b) $P(A \cap B) \neq 0$

(d) $P(A|B) = P(A)$ (b) 16 (c) 2
18. Which of the following statements regarding probability are correct?

i. $0 \leq P(A) \leq 1$ for any event A

ii. $P(S) = 1$ where S is the sample space

iii. For mutually exclusive events A and B , $P(A \cup B) = P(A) + P(B)$

Which one is correct?

(a) i and ii (b) i and iii
23. If $Y = 3X + 5$ and $V(X) = 4$, what is the variance $V(Y)$?

(a) 17 (b) 36 (c) 4
24. For two independent random variables X and Y , which of the following properties are true?

(c) ii and iii (d) i, ii and iii

i. $E(X + Y) = E(X) + E(Y)$

ii. $E(XY) = E(X)E(Y)$

iii. $V(X - Y) = V(X) + V(Y)$

Which one is correct?

(a) i and ii (b) i and iii (c) ii
- Answer the next TWO questions based on the following information

In a group of 100 students, 60 passed in Mathematics, 50 passed in Statistics, and 30 passed in both subjects. A student is selected at random.
19. What is the probability that the student passed in Mathematics or Statistics?

(a) 0.80 (b) 0.50
25. If $E(X) = 5$, what is the value of $E(2X - 3)$?

(c) 0.30 (d) 10 (c) 1

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Candidates are asked not to leave any mark or spot on the question paper.

1. The first central moment (μ_1) of any frequency distribution is always equal to –
(a) 1 (b) 0 (c) -1 (d) Mean
Which one is correct?

2. Which moment is used to measure the variance (dispersion) of a distribution?
(a) First raw moment (μ'_1)
(c) Second central moment (μ_2)

3. For a mesokurtic (normal-shaped) distribution, what is the value of kurtosis β_2 ?
(a) 0 (b) 1 (c) 3 (d) -3

4. Which of the following statements about moments are correct?
i. The first raw moment around zero (μ'_1) is the arithmetic mean.
ii. The third central moment (μ_3) can be negative, zero, or positive.
iii. The second central moment (μ_2) can never be negative.
Which one is correct?
(a) i and ii (b) i and iii (c) ii and iii (d) i, ii and iii

5. If the first raw moment about 4 is -2 , what is the value of the arithmetic mean?
(a) 2 (b) -2 (c) 6 (d) 4

6. What is the probability of a certain event (an event guaranteed to occur)?
(a) 0 (b) 0.5 (c) 1 (d) Undefined

7. If two events A and B are mutually exclusive (disjoint), which of the following is true?
(a) $P(A \cap B) = P(A)P(B)$
(c) $P(A \cup B) = 1$
(b) $P(A \cap B) = 0$
(d) $P(A \cup B) = P(A) + P(B)$
Which one is correct?
(a) i and ii (b) i and iii (c) ii and iii (d) i, ii and iii

8. Which of the following statements regarding probability are correct?
i. $0 \leq P(A) \leq 1$ for any event A
ii. $P(S) = 1$ where S is the sample space

9. What is the probability that the student passed in Mathematics or Statistics?
(a) 0.80 (b) 0.50 (c) 0.30 (d) 0.70

10. What is the conditional probability that the student passed in Statistics given that s/he passed in Mathematics?
(a) 0.30 (b) 0.50 (c) 0.70 (d) 0.20

11. If X is a random variable and c is a constant, what is the variance $V(c)$?
(a) c (b) c^2 (c) 0 (d) $c^2 + 1$

12. If $E(X) = 4$ and $V(X) = 5$, what is the value of $E(X^2)$?
(a) 9 (b) 16 (c) 29 (d) 41

13. If $Y = 3X + 5$ and $V(X) = 4$, what is the variance $V(Y)$?
(a) 17 (b) 36 (c) 49 (d) 61

14. For two independent random variables X and Y , which of the following properties are true?
i. $E(X + Y) = E(X) + E(Y)$
ii. $E(XY) = E(X)E(Y)$
iii. $V(X + Y) = V(X) + V(Y)$
iv. $P(A \cap B) = P(A)P(B)$
Which one is correct?
(a) i and ii (b) i and iii (c) ii and iii (d) i, ii and iii

15. If $E(X) = 5$, what is the value of $E(2X - 3)$?
(a) 7 (b) 10 (c) 13 (d) 16

16. Which of the following is an example of a continuous random variable?

(a) 0.25
(b) 0.50
(c) 0

(a) Number of defective items in a batch
(b) Height of students in a class

(c) Number of phone calls received in an hour
(d) Variables lying within which range?

(a) $0 \leq r \leq 1$
(b) $-1 \leq r \leq 1$
(c) -
17. What is $F(\infty)$ for any valid cumulative distribution function (CDF) $F(x)$?

(a) $-\infty$
(b) -1
18. Which of the following are necessary conditions for a discrete probability function $P(x)$?

i. $P(x_i) \geq 0$ for all x_i
ii. $\sum P(x_i) = 1$
iii. $P(x_i) \leq 1$ for all x_i

Which one is correct?

(a) i and ii
(b) i and iii
- Answer the next TWO questions based on the following information

A discrete random variable X has the following probability distribution:

X	0	1	2
$P(x)$	0.50	0.25	0.25
19. What is the value of the cumulative distribution function $F(1)$?

(a) 0.25
(b) 0.50
20. What is $P(X \geq 1)$?
21. The correlation coefficient r between two variables lies within which range?

(a) 0.6
(b) 1.2
(c) 0
22. If the correlation coefficient between X and Y is $r = 0.6$, what is the correlation coefficient between $2X + 3$ and $3Y - 5$?

(a) 0.6
(b) 1.2
(c) 0
23. The correlation coefficient r is the geometric mean of –

(a) Two means
(b) T

(c) Two regression coefficients (b_{yx} and b_{xy})
(d) T

(c) ii and iii
(d) i, ii and iii
24. Which of the following statements are true about the correlation coefficient r ?

i. It is independent of change of origin and scale.
ii. It is a unitless measure.
iii. Correlation implies a causal relationship between variables.

Which one is correct?

(a) i and ii
(b) i and iii
(c) ii
25. If the two regression coefficients are $b_{yx} = 0.8$ and $b_{xy} = 0.45$, what is the value of the correlation coefficient r ?

(a) 0.75
(b) 1.00

(a) 0.32
(b) 0.60
(c) 0